



CLOUDFLARE RADAR • 2024 YEAR IN REVIEW

Cloudflare Radar 2024 Year in Review

A data-driven executive view of Internet traffic, technology adoption, connectivity disruptions, and security threats observed through Cloudflare's global network.

Audience: technology & network leaders

Scope: global Internet trends

Source: Cloudflare Radar

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Year-over-year global Internet traffic growth in 2024.

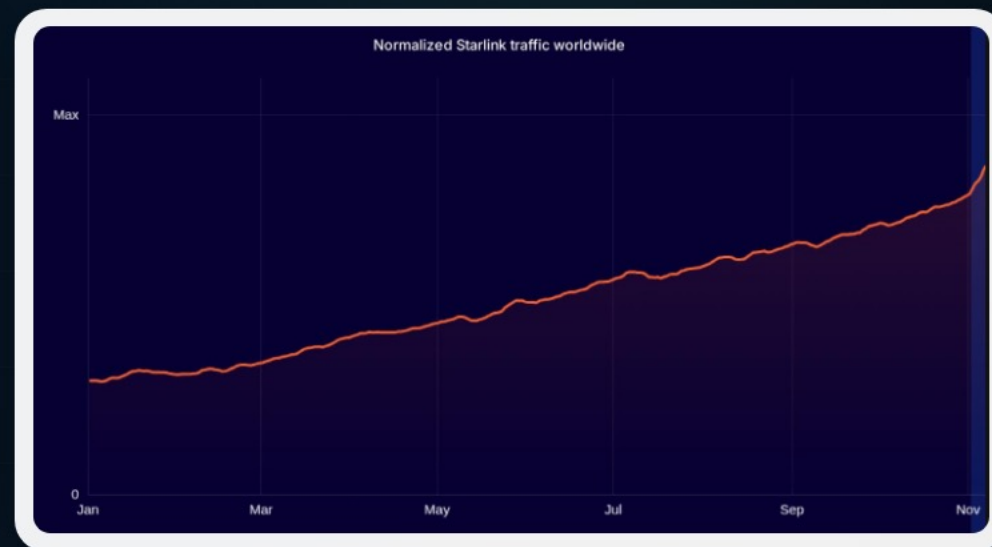
3.3x

SpaceX Starlink traffic growth during 2024.

Mid-Jan

Trend line starts mid-January to normalize for post-holiday activity.

Executive signal: baseline Internet demand is still compounding, while satellite connectivity is expanding reach into previously underserved locations.



Google Leads Top 10 Services as Generative AI Rises

#1 Google

#2 Facebook

#3 Apple

#4 TikTok

#5 AWS

#6 Microsoft

#7 Instagram

#8 YouTube

#9 Amazon

#10 WhatsApp

Market shift: generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background.

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

2% → 13%

Post-quantum encryption share of TLS 1.3 traffic in 2024.

21%

Traffic using HTTP/3, running on QUIC with encryption by default.

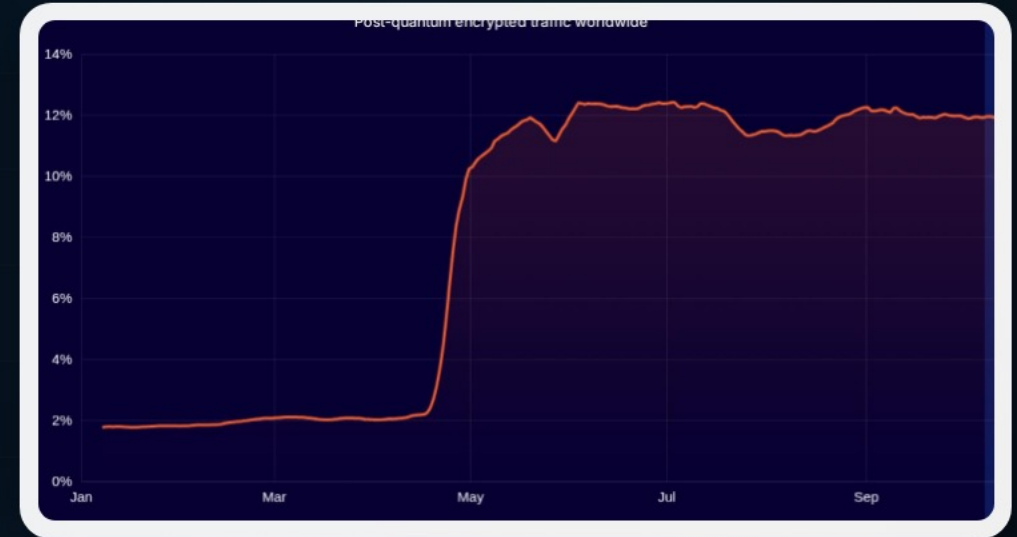
Go

Most popular API client language, at roughly 12% of automated API requests.

Apr 17

Chrome 124 enabled post-quantum key agreement by default.

Firefox has started rolling out post-quantum by default, and Safari began initial testing.



225 Major Disruptions: Government Shutdowns Lead Outage Causes

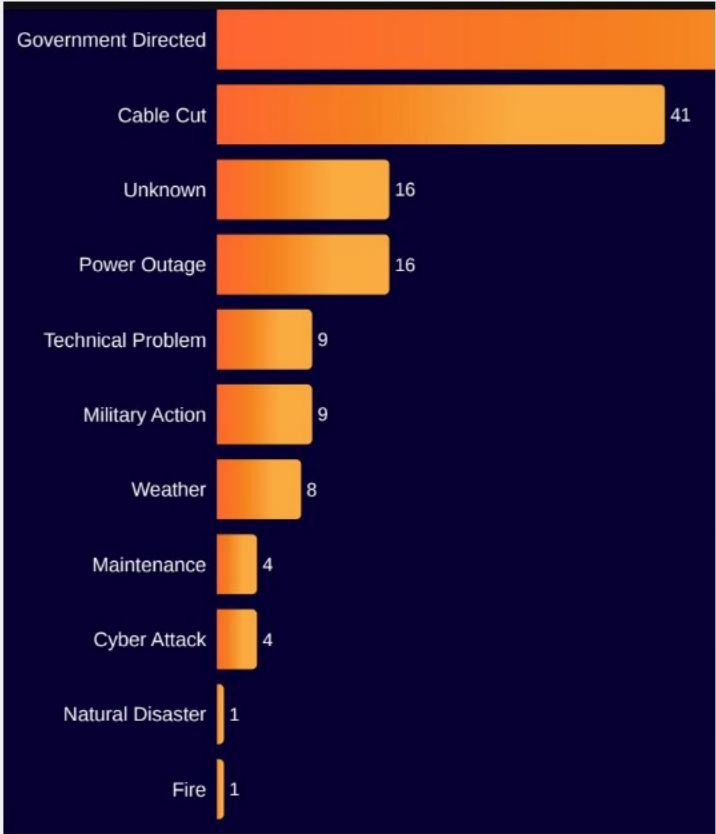
225

Major Internet disruptions observed globally in 2024.

116+

Government-directed disruptions, the leading cause.

CAUSE	COUNT
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance / Cyber Attack	4 each
Natural Disaster / Fire	1 each



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Share of all Cloudflare-processed traffic that was mitigated.

Gambling / Games

Most attacked industry category in 2024.

68%

Global bot traffic originating from the top 10 countries.

Log4j

CVE-2021-44228 remains actively targeted more than two years after disclosure.

Email Threat Metrics

Malicious emails processed

4.3%

Malicious emails with deceptive links

43%

The top-level domain **.bar** originated the largest share of malicious and spam email observed during the year.

Security Posture Signal

+6.4%

Increase in RPKI valid routes.

Persistent

Known vulnerabilities continue to shape attacker behavior.

Five Takeaways Shaping the 2025 Internet Outlook

01

Traffic keeps compounding

Global Internet traffic grew 17% year over year, with Starlink traffic surging 3.3x.

02

Quantum readiness accelerates

Post-quantum encryption moved from roughly 2% to 13% of TLS 1.3 traffic after Chrome 124 default enablement.

03

Connectivity fragility is geopolitical

Government-directed shutdowns led the 225 major disruptions observed, ahead of cable cuts and power outages.

04

Threat pressure remains pervasive

Cloudflare mitigated 6.5% of traffic, Gambling/Games was most targeted, and Log4j exploitation persisted.

05

Developer ecosystems shift

Go became the most popular API client language, capturing roughly 12% of automated API traffic.